```
console.log("A ", myArr);

const myn1 = myArr.slice(1, 3)

console.log(myn1);
console.log("B ", myArr);


const myn2 = myArr.splice(1, 3)
console.log("C ", myArr);
console.log(myn2);



function sayMyName(){
    console.log("H");
    console.log("I");
    console.log("T");
    console.log("E");
    console.log("S");
    console.log("H");
}

// sayMyName()

// function addTwoNumbers(number1, number2){

//     console.log(number1 + number2);
// }

function addTwoNumbers(number1, number2){
```

```
    // let result = number1 + number2
    // return result
    return number1 + number2
}

const result = addTwoNumbers(3, 5)

// console.log("Result: ", result);


function loginUserMessage(username = "sam"){
    if(!username){
        console.log("PLease enter a username");
        return
    }
    return `${username} just logged in`
}

// console.log(loginUserMessage("hitesh"))
// console.log(loginUserMessage("hitesh"))


function calculateCartPrice(val1, val2, ...num1){
    return num1
}

// console.log(calculateCartPrice(200, 400, 500, 2000))

const user = {
    username: "hitesh",
    prices: 199
}
```

```
function handleObject(anyobject){
    console.log(`Username is ${anyobject.username} and price is ${anyobject.price}`);
}

// handleObject(user)
handleObject({
    username: "sam",
    price: 399
})

const myNewArray = [200, 400, 100, 600]

function returnSecondValue(getArray){
    return getArray[1]
}

// console.log(returnSecondValue(myNewArray));
console.log(returnSecondValue([200, 400, 500, 1000]));

const accountId = 144553
let accountEmail = "hitesh@google.com"
var accountPassword = "12345"
accountCity = "Jaipur"
let accountState;

// accountId = 2 // not allowed


accountEmail = "hc@hc.com"
accountPassword = "21212121"
accountCity = "Bengaluru"
```

```
console.log(accountId);

/*
Prefer not to use var
because of issue in block scope and functional scope
*/


console.table([accountId, accountEmail, accountPassword, accountCity, accountState])


const marvel_heros = ["thor", "Ironman", "spiderman"]
const dc_heros = ["superman", "flash", "batman"]

// marvel_heros.push(dc_heros)

// console.log(marvel_heros);
// console.log(marvel_heros[3][1]);

// const allHeros = marvel_heros.concat(dc_heros)
// console.log(allHeros);

const all_new_heros = [...marvel_heros, ...dc_heros]

// console.log(all_new_heros);

const another_array = [1, 2, 3, [4, 5, 6], 7, [6, 7, [4, 5]]]

const real_another_array = another_array.flat(Infinity)
console.log(real_another_array);
```

```
console.log(Array.isArray("Hitesh"))
console.log(Array.from("Hitesh"))
console.log(Array.from({name: "hitesh"})) // interesting

let score1 = 100
let score2 = 200
let score3 = 300

console.log(Array.of(score1, score2, score3));


"use strict"; // treat all JS code as newer version

// alert( 3 + 3) // we are using nodejs, not browser

console.log(3
    +
     3) // code readability should be high

console.log("Hitesh")


let name = "hitesh"
let age = 18
let isLoggedIn = false
let state;

// number => 2 to power 53
// bigint
// string => ""
```

```
// boolean => true/false
// null => standalone value
// undefined =>
// symbol => unique


// object

console.log(typeof undefined); // undefined
console.log(typeof null); // object


//var c = 300
let a = 300
if (true) {
    let a = 10
    const b = 20
    // console.log("INNER: ", a);

}



// console.log(a);
// console.log(b);
// console.log(c);


function one(){
    const username = "hitesh"

    function two(){
```

```
            const website = "youtube"
            console.log(username);
        }
        // console.log(website);

         two()

}

// one()

if (true) {
        const username = "hitesh"
        if (username === "hitesh") {
            const website = " youtube"
            // console.log(username + website);
        }
        // console.log(website);
}

// console.log(username);


// ++++++++++++++++++ interesting ++++++++++++++++++


console.log(addone(5))

function addone(num){
        return num + 1
}
```

```
addTwo(5)
const addTwo = function(num){
    return num + 2
}


const user = {
    username: "hitesh",
    price: 999,

    welcomeMessage: function() {
        console.log(`${this.username} , welcome to website`);
        console.log(this);
    }

}

// user.welcomeMessage()
// user.username = "sam"
// user.welcomeMessage()

// console.log(this);

// function chai(){
//     let username = "hitesh"
//     console.log(this.username);
// }

// chai()
```

```
// const chai = function () {
//     let username = "hitesh"
//     console.log(this.username);
// }

const chai =  () => {
    let username = "hitesh"
    console.log(this);
}


// chai()

// const addTwo = (num1, num2) => {
//     return num1 + num2
// }

// const addTwo = (num1, num2) =>  num1 + num2

// const addTwo = (num1, num2) => ( num1 + num2 )

const addTwo = (num1, num2) => ({username: "hitesh"})


console.log(addTwo(3, 4))


// const myArray = [2, 5, 3, 7, 8]

// myArray.forEach()
```

```
let score = "hitesh"

//console.log(typeof score);
//console.log(typeof(score));

let valueInNumber = Number(score)
//console.log(typeof valueInNumber);
//console.log(valueInNumber);



// "33" => 33
// "33abc" => NaN
//   true => 1; false => 0

let isLoggedIn = "hitesh"

let booleanIsLoggedIn = Boolean(isLoggedIn)
//   console.log(booleanIsLoggedIn);

// 1 => true; 0 => false
// "" => false
// "hitesh" => true

let someNumber = 33

let stringNumber = String(someNumber)
// console.log(stringNumber);
// console.log(typeof stringNumber);

// ********** Operations **********

let value = 3
```

```
let negValue = -value
// console.log(negValue);

// console.log(2+2);
// console.log(2-2);
// console.log(2*2);
// console.log(2**3);
// console.log(2/3);
// console.log(2%3);

let str1 = "hello"
let str2 = " hitesh"

let str3 = str1 + str2
// console.log(str3);

// console.log("1" + 2);
// console.log(1 + "2");
// console.log("1" + 2 + 2);
// console.log(1 + 2 + "2");

// console.log( (3 + 4) * 5 % 3);

// console.log(+true);
// console.log(+"");

let num1, num2, num3

num1 = num2 = num3 = 2 + 2

let gameCounter = 100
++gameCounter;
```

```
console.log(gameCounter);

// link to study
// https://tc39.es/ecma262/multipage/abstract-operations.html#sec-type-conversion



// singleton
// Object.create

// object literals

const mySym = Symbol("key1")


const JsUser = {
    name: "Hitesh",
    "full name": "Hitesh Choudhary",
    [mySym]: "mykey1",
    age: 18,
    location: "Jaipur",
    email: "hitesh@google.com",
    isLoggedIn: false,
    lastLoginDays: ["Monday", "Saturday"]
}

// console.log(JsUser.email)
// console.log(JsUser["email"])
// console.log(JsUser["full name"])
// console.log(JsUser[mySym])

JsUser.email = "hitesh@chatgpt.com"
// Object.freeze(JsUser)
```

```
JsUser.email = "hitesh@microsoft.com"
// console.log(JsUser);

JsUser.greeting = function(){
    console.log("Hello JS user");
}
JsUser.greetingTwo = function(){
    console.log(`Hello JS user, ${this.name}`);
}

console.log(JsUser.greeting());
console.log(JsUser.greetingTwo());


// console.log(2 > 1);
// console.log(2 >= 1);
// console.log(2 < 1);
// console.log(2 == 1);
// console.log(2 != 1);


// console.log("2" > 1);
// console.log("02" > 1);

console.log(null > 0);
console.log(null == 0);
console.log(null >= 0);

console.log(undefined == 0);
console.log(undefined > 0);
console.log(undefined < 0);
```

```
// ===

console.log("2" === 2);


// Immediately Invoked Function Expressions (IIFE)


(function chai(){
    // named IIFE
    console.log(`DB CONNECTED`);
})();

( (name) => {
    console.log(`DB CONNECTED TWO ${name}`);
} )('hitesh')


// const tinderUser = new Object()
const tinderUser = {}

tinderUser.id = "123abc"
tinderUser.name = "Sammy"
tinderUser.isLoggedIn = false

// console.log(tinderUser);

const regularUser = {
    email: "some@gmail.com",
    fullname: {
        userfullname: {
            firstname: "hitesh",
```

```
            lastname: "choudhary"
        }
    }
}

// console.log(regularUser.fullname.userfullname.firstname);

const obj1 = {1: "a", 2: "b"}
const obj2 = {3: "a", 4: "b"}
const obj4 = {5: "a", 6: "b"}

// const obj3 = { obj1, obj2 }
// const obj3 = Object.assign({}, obj1, obj2, obj4)

const obj3 = {...obj1, ...obj2}
// console.log(obj3);



const users = [
    {
        id: 1,
        email: "h@gmail.com"
    },
    {
        id: 1,
        email: "h@gmail.com"
    },
    {
        id: 1,
        email: "h@gmail.com"
    },
]
```

```
users[1].email
// console.log(tinderUser);

// console.log(Object.keys(tinderUser));
// console.log(Object.values(tinderUser));
// console.log(Object.entries(tinderUser));

// console.log(tinderUser.hasOwnProperty('isLoggedIn'));


const course = {
    coursename: "js in hindi",
    price: "999",
    courseInstructor: "hitesh"
}

// course.courseInstructor

const {courseInstructor: instructor} = course

// console.log(courseInstructor);
console.log(instructor);

// {
//     "name": "hitesh",
//     "coursename": "js in hindi",
//     "price": "free"
// }

[
    {},
```

```
    {},
    {}
]

const name = "hitesh"
const repoCount = 50

// console.log(name + repoCount + " Value");

console.log(`Hello my name is ${name} and my repo count is ${repoCount}`);

const gameName = new String('hitesh-hc-com')

// console.log(gameName[0]);
// console.log(gameName.__proto__);



// console.log(gameName.length);
// console.log(gameName.toUpperCase());
console.log(gameName.charAt(2));
console.log(gameName.indexOf('t'));

const newString = gameName.substring(0, 4)
console.log(newString);

const anotherString = gameName.slice(-8, 4)
console.log(anotherString);

const newStringOne = "    hitesh      "
console.log(newStringOne);
console.log(newStringOne.trim());
```

```
const url = "https://hitesh.com/hitesh%20choudhary"

console.log(url.replace('%20', '-'))

console.log(url.includes('sundar'))

console.log(gameName.split('-'));


const score = 400
// console.log(score);

const balance = new Number(100)
// console.log(balance);

// console.log(balance.toString().length);
// console.log(balance.toFixed(1));

const otherNumber = 123.8966

// console.log(otherNumber.toPrecision(4));

const hundreds = 1000000
// console.log(hundreds.toLocaleString('en-IN'));

// ++++++++++++++ Maths +++++++++++++++++++++++++++++

// console.log(Math);
// console.log(Math.abs(-4));
// console.log(Math.round(4.6));
// console.log(Math.ceil(4.2));
// console.log(Math.floor(4.9));
```

```
// console.log(Math.min(4, 3, 6, 8));
// console.log(Math.max(4, 3, 6, 8));

console.log(Math.random());
console.log((Math.random()*10) + 1);
console.log(Math.floor(Math.random()*10) + 1);

const min = 10
const max = 20

console.log(Math.floor(Math.random() * (max - min + 1)) + min)

// Dates

let myDate = new Date()
// console.log(myDate.toString());
// console.log(myDate.toDateString());
// console.log(myDate.toLocaleString());
// console.log(typeof myDate);

// let myCreatedDate = new Date(2023, 0, 23)
// let myCreatedDate = new Date(2023, 0, 23, 5, 3)
// let myCreatedDate = new Date("2023-01-14")
let myCreatedDate = new Date("01-14-2023")
// console.log(myCreatedDate.toLocaleString());

let myTimeStamp = Date.now()

// console.log(myTimeStamp);
// console.log(myCreatedDate.getTime());
// console.log(Math.floor(Date.now()/1000));
```

```
let newDate = new Date()
console.log(newDate);
console.log(newDate.getMonth() + 1);
console.log(newDate.getDay());

// `${newDate.getDay()} and the time `

newDate.toLocaleString('default', {
      weekday: "long",

})



//   Primitive

//   7 types : String, Number, Boolearn, null, undefined, Symbol, BigInt

const score = 100
const scoreValue = 100.3

const isLoggedIn = false
const outsideTemp = null
let userEmail;

const id = Symbol('123')
const anotherId = Symbol('123')

console.log(id === anotherId);

// const bigNumber = 3456543576654356754n
```

```
// Reference (Non primitive)

// Array, Objects, Functions

const heros = ["shaktiman", "naagraj", "doga"];
let myObj = {
    name: "hitesh",
    age: 22,
}

const myFunction = function(){
    console.log("Hello world");
}

console.log(typeof anotherId);

// https://262.ecma-international.org/5.1/#sec-11.4.3


const coding = ["js", "ruby", "java", "python", "cpp"]

// coding.forEach( function (val){
//     console.log(val);
// } )

// coding.forEach( (item) => {
//     console.log(item);
// } )

// function printMe(item){
//     console.log(item);
```

```
// }

// coding.forEach(printMe)

// coding.forEach( (item, index, arr)=> {
//       console.log(item, index, arr);
// } )

const myCoding = [
    {
        languageName: "javascript",
        languageFileName: "js"
    },
    {
        languageName: "java",
        languageFileName: "java"
    },
    {
        languageName: "python",
        languageFileName: "py"
    },
]

myCoding.forEach( (item) => {

    console.log(item.languageName);
} )


const myObject = {
    js: 'javascript',
    cpp: 'C++',
```

```
    rb: "ruby",
    swift: "swift by apple"
}

for (const key in myObject) {
    //console.log(`${key} shortcut is for ${myObject[key]}`);
}

const programming = ["js", "rb", "py", "java", "cpp"]

for (const key in programming) {
    //console.log(programming[key]);
}

// const map = new Map()
// map.set('IN', "India")
// map.set('USA', "United States of America")
// map.set('Fr', "France")
// map.set('IN', "India")

// for (const key in map) {
//     console.log(key);
// }



const myNums = [1, 2, 3]

// const myTotal = myNums.reduce(function (acc, currval) {
//     console.log(`acc: ${acc} and currval: ${currval}`);
//     return acc + currval
// }, 0)
```

```
const myTotal = myNums.reduce( (acc, curr) => acc+curr, 0)

console.log(myTotal);


const shoppingCart = [
    {
        itemName: "js course",
        price: 2999
    },
    {
        itemName: "py course",
        price: 999
    },
    {
        itemName: "mobile dev course",
        price: 5999
    },
    {
        itemName: "data science course",
        price: 12999
    },
]

const priceToPay = shoppingCart.reduce((acc, item) => acc + item.price, 0)

console.log(priceToPay);


// for
```

```
for (let i = 0; i <= 10; i++) {
      const element = i;
      if (element == 5) {
            //console.log("5 is best number");
      }
      //console.log(element);

}

// console.log(element);

for (let i = 1; i <= 10; i++) {
      //console.log(`Outer loop value: ${i}`);
    for (let j = 1; j <= 10; j++) {
      //console.log(`Inner loop value ${j} and inner loop ${i}`);
      //console.log(i + '*' + j + ' = ' + i*j );
    }

}
let myArray = ["flash", "batman", "superman"]
//console.log(myArray.length);
for (let index = 0; index < myArray.length; index++) {
      const element = myArray[index];
      //console.log(element);

}


// break and continue

// for (let index = 1; index <= 20; index++) {
//       if (index == 5) {
```

```
//              console.log(`Detected 5`);
//              break
//       }
//      console.log(`Value of i is ${index}`);

// }

for (let index = 1; index <= 20; index++) {
      if (index == 5) {
            console.log(`Detected 5`);
            continue
      }
     console.log(`Value of i is ${index}`);

}


const myNumers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

// const newNums = myNumers.map( (num) => { return num + 10})

const newNums = myNumers
                     .map((num) => num * 10 )
                     .map( (num) => num + 1)
                     .filter( (num) => num >= 40)

console.log(newNums);


// const coding = ["js", "ruby", "java", "python", "cpp"]
```

```
// const values = coding.forEach( (item) => {
//       //console.log(item);
//       return item
// } )

// console.log(values);

const myNums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

// const newNums = myNums.filter( (num) => {
//       return num > 4
// } )

// const newNums = []

// myNums.forEach( (num) => {
//       if (num > 4) {
//             newNums.push(num)
//       }
// } )

// console.log(newNums);


const books = [
      { title: 'Book One', genre: 'Fiction', publish: 1981, edition: 2004 },
      { title: 'Book Two', genre: 'Non-Fiction', publish: 1992, edition: 2008 },
      { title: 'Book Three', genre: 'History', publish: 1999, edition: 2007 },
      { title: 'Book Four', genre: 'Non-Fiction', publish: 1989, edition: 2010 },
      { title: 'Book Five', genre: 'Science', publish: 2009, edition: 2014 },
      { title: 'Book Six', genre: 'Fiction', publish: 1987, edition: 2010 },
      { title: 'Book Seven', genre: 'History', publish: 1986, edition: 1996 },
```

```
    { title: 'Book Eight', genre: 'Science', publish: 2011, edition: 2016 },
    { title: 'Book Nine', genre: 'Non-Fiction', publish: 1981, edition: 1989 },
  ];

  let userBooks = books.filter( (bk) => bk.genre === 'History')

  userBooks = books.filter( (bk) => {
    return bk.publish >= 1995 && bk.genre === "History"
})
  console.log(userBooks);



  // switch (key) {
//       case value:

//           break;

//       default:
//           break;
// }

const month = "march"

switch (month) {
    case "jan":
        console.log("January");
        break;
    case "feb":
        console.log("feb");
        break;
    case "march":
        console.log("march");
```

```
            break;
        case "april":
            console.log("april");
            break;

        default:
            console.log("default case match");
            break;
}

// for of

// ["", "", ""]
// [{}, {}, {}]

const arr = [1, 2, 3, 4, 5]

for (const num of arr) {
    //console.log(num);
}

const greetings = "Hello world!"
for (const greet of greetings) {
    //console.log(`Each char is ${greet}`)
}

// Maps

const map = new Map()
map.set('IN', "India")
map.set('USA', "United States of America")
map.set('Fr', "France")
```

```
map.set('IN', "India")


// console.log(map);

for (const [key, value] of map) {
    // console.log(key, ':-', value);
}

const myObject = {
    game1: 'NFS',
    game2: 'Spiderman'
}

// for (const [key, value] of myObject) {
//     console.log(key, ':-', value);

// }


const userEmail = []

if (userEmail) {
    console.log("Got user email");
} else {
    console.log("Don't have user email");
}

// falsy values

// false, 0, -0, BigInt 0n, "", null, undefined, NaN
```

```
//truthy values
// "0", 'false', " ", [], {}, function(){}

// if (userEmail.length === 0) {
//       console.log("Array is empty");
// }

const emptyObj = {}

if (Object.keys(emptyObj).length === 0) {
      console.log("Object is empty");
}

// Nullish Coalescing Operator (??): null undefined

let val1;
// val1 = 5 ?? 10
// val1 = null ?? 10
// val1 = undefined ?? 15
val1 = null ?? 10 ?? 20



console.log(val1);

// Terniary Operator

// condition ? true : false

const iceTeaPrice = 100
iceTeaPrice <= 80 ? console.log("less than 80") : console.log("more than 80")
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta http-equiv="X-UA-Compatible" content="IE=edge">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>DOM | Chai aur code</title>
</head>
<body style="background-color: #212121; color: #fff;">
    <div class="parent">
        <!-- this is a comment -->
        <div class="day">Monday</div>
        <div class="day">Tuesday</div>
        <div class="day">Wednesday</div>
        <div class="day">Thursday</div>
    </div>
</body>
<script>
    const parent = document.querySelector('.parent')
    // console.log(parent);
    // console.log(parent.children);
    // console.log(parent.children[1].innerHTML);

    // for (let i = 0; i < parent.children.length; i++) {
    //         console.log(parent.children[i].innerHTML);

    // }
    parent.children[1].style.color = "orange"
    // console.log(parent.firstElementChild);
    // console.log(parent.lastElementChild);
```

```
    const dayOne = document.querySelector('.day')
    // console.log(dayOne);
    // console.log(dayOne.parentElement);
    // console.log(dayOne.nextElementSibling);

    console.log("NODES: ", parent.childNodes);
</script>
</html>



let index = 0
// while (index <= 10) {
//     console.log(`Value of index is ${index}`);
//     index = index + 2
// }

let myArray = ['flash', "batman", "superman"]

let arr = 0
while (arr < myArray.length) {
    //console.log(`Value is ${myArray[arr]}`);
    arr = arr + 1
}

let score = 11

do {
    console.log(`Score is ${score}`);
    score++
} while (score <= 10);
```

```
// array

const myArr = [0, 1, 2, 3, 4, 5]
const myHeors = ["shaktiman", "naagraj"]

const myArr2 = new Array(1, 2, 3, 4)
// console.log(myArr[1]);

// Array methods

// myArr.push(6)
// myArr.push(7)
// myArr.pop()

// myArr.unshift(9)
// myArr.shift()

// console.log(myArr.includes(9));
// console.log(myArr.indexOf(3));

// const newArr = myArr.join()

// console.log(myArr);
// console.log( newArr);


// slice, splice

console.log("A ", myArr);

const myn1 = myArr.slice(1, 3)
```

```
console.log(myn1);
console.log("B ", myArr);


const myn2 = myArr.splice(1, 3)
console.log("C ", myArr);
console.log(myn2);


const marvel_heros = ["thor", "Ironman", "spiderman"]
const dc_heros = ["superman", "flash", "batman"]

// marvel_heros.push(dc_heros)

// console.log(marvel_heros);
// console.log(marvel_heros[3][1]);

// const allHeros = marvel_heros.concat(dc_heros)
// console.log(allHeros);

const all_new_heros = [...marvel_heros, ...dc_heros]

// console.log(all_new_heros);

const another_array = [1, 2, 3, [4, 5, 6], 7, [6, 7, [4, 5]]]

const real_another_array = another_array.flat(Infinity)
console.log(real_another_array);
```

```
console.log(Array.isArray("Hitesh"))
console.log(Array.from("Hitesh"))
console.log(Array.from({name: "hitesh"})) // interesting

let score1 = 100
let score2 = 200
let score3 = 300

console.log(Array.of(score1, score2, score3));



// singleton
// Object.create

// object literals

const mySym = Symbol("key1")


const JsUser = {
    name: "Hitesh",
    "full name": "Hitesh Choudhary",
    [mySym]: "mykey1",
    age: 18,
    location: "Jaipur",
    email: "hitesh@google.com",
    isLoggedIn: false,
    lastLoginDays: ["Monday", "Saturday"]
}

// console.log(JsUser.email)
```

```
// console.log(JsUser["email"])
// console.log(JsUser["full name"])
// console.log(JsUser[mySym])

JsUser.email = "hitesh@chatgpt.com"
// Object.freeze(JsUser)
JsUser.email = "hitesh@microsoft.com"
// console.log(JsUser);

JsUser.greeting = function(){
    console.log("Hello JS user");
}
JsUser.greetingTwo = function(){
    console.log(`Hello JS user, ${this.name}`);
}

console.log(JsUser.greeting());
console.log(JsUser.greetingTwo());
// const tinderUser = new Object()
const tinderUser = {}

tinderUser.id = "123abc"
tinderUser.name = "Sammy"
tinderUser.isLoggedIn = false

// console.log(tinderUser);

const regularUser = {
    email: "some@gmail.com",
    fullname: {
        userfullname: {
            firstname: "hitesh",
```

```
            lastname: "choudhary"
        }
    }
}

// console.log(regularUser.fullname.userfullname.firstname);

const obj1 = {1: "a", 2: "b"}
const obj2 = {3: "a", 4: "b"}
const obj4 = {5: "a", 6: "b"}

// const obj3 = { obj1, obj2 }
// const obj3 = Object.assign({}, obj1, obj2, obj4)

const obj3 = {...obj1, ...obj2}
// console.log(obj3);



const users = [
    {
        id: 1,
        email: "h@gmail.com"
    },
    {
        id: 1,
        email: "h@gmail.com"
    },
    {
        id: 1,
        email: "h@gmail.com"
    },
]
```

```
users[1].email
// console.log(tinderUser);

// console.log(Object.keys(tinderUser));
// console.log(Object.values(tinderUser));
// console.log(Object.entries(tinderUser));

// console.log(tinderUser.hasOwnProperty('isLoggedIn'));


const course = {
    coursename: "js in hindi",
    price: "999",
    courseInstructor: "hitesh"
}

// course.courseInstructor

const {courseInstructor: instructor} = course

// console.log(courseInstructor);
console.log(instructor);

// {
//     "name": "hitesh",
//     "coursename": "js in hindi",
//     "price": "free"
// }

[
    {},
```

```
    {},
    {}
]



function sayMyName(){
    console.log("H");
    console.log("I");
    console.log("T");
    console.log("E");
    console.log("S");
    console.log("H");
}

// sayMyName()

// function addTwoNumbers(number1, number2){

//     console.log(number1 + number2);
// }

function addTwoNumbers(number1, number2){

    // let result = number1 + number2
    // return result
    return number1 + number2
}

const result = addTwoNumbers(3, 5)

// console.log("Result: ", result);
```

```
function loginUserMessage(username = "sam"){
    if(!username){
        console.log("PLease enter a username");
        return
    }
    return `${username} just logged in`
}

// console.log(loginUserMessage("hitesh"))
// console.log(loginUserMessage("hitesh"))


function calculateCartPrice(val1, val2, ...num1){
    return num1
}

// console.log(calculateCartPrice(200, 400, 500, 2000))

const user = {
    username: "hitesh",
    prices: 199
}

function handleObject(anyobject){
    console.log(`Username is ${anyobject.username} and price is ${anyobject.price}`);
}

// handleObject(user)
handleObject({
    username: "sam",
```

```
    price: 399
})

const myNewArray = [200, 400, 100, 600]

function returnSecondValue(getArray){
    return getArray[1]
}

// console.log(returnSecondValue(myNewArray));
console.log(returnSecondValue([200, 400, 500, 1000]));


//var c = 300
let a = 300
if (true) {
    let a = 10
    const b = 20
    // console.log("INNER: ", a);

}



// console.log(a);
// console.log(b);
// console.log(c);


function one(){
    const username = "hitesh"
```

```
    function two(){
        const website = "youtube"
        console.log(username);
    }
    // console.log(website);

     two()

}

// one()

if (true) {
    const username = "hitesh"
    if (username === "hitesh") {
        const website = " youtube"
        // console.log(username + website);
    }
    // console.log(website);
}

// console.log(username);


// ++++++++++++++++++ interesting ++++++++++++++++++


console.log(addone(5))

function addone(num){
    return num + 1
}
```

```
addTwo(5)
const addTwo = function(num){
    return num + 2
}


const user = {
    username: "hitesh",
    price: 999,

    welcomeMessage: function() {
        console.log(`${this.username} , welcome to website`);
        console.log(this);
    }

}

// user.welcomeMessage()
// user.username = "sam"
// user.welcomeMessage()

// console.log(this);

// function chai(){
//     let username = "hitesh"
//     console.log(this.username);
// }

// chai()
```

```
// const chai = function () {
//     let username = "hitesh"
//     console.log(this.username);
// }

const chai =  () => {
    let username = "hitesh"
    console.log(this);
}


// chai()

// const addTwo = (num1, num2) => {
//     return num1 + num2
// }

// const addTwo = (num1, num2) =>  num1 + num2

// const addTwo = (num1, num2) => ( num1 + num2 )

const addTwo = (num1, num2) => ({username: "hitesh"})


console.log(addTwo(3, 4))


// const myArray = [2, 5, 3, 7, 8]

// myArray.forEach()
```

```
// Immediately Invoked Function Expressions (IIFE)


(function chai(){
    // named IIFE
    console.log(`DB CONNECTED`);
})();

( (name) => {
    console.log(`DB CONNECTED TWO ${name}`);
} )('hitesh')

// if
const isUserloggedIn = true
const temperature = 41

// if ( temperature === 40 ){
//     console.log("less than 50");
// } else {
//     console.log("temperature is greater than 50");
// }

// console.log("Execute");
// <, >, <=, >=, ==, !=, ===, !==

// const score = 200

// if (score > 100) {
//     let power = "fly"
//     console.log(`User power: ${power}`);
// }
```

```
// console.log(`User power: ${power}`);


// const balance = 1000

// if (balance > 500) console.log("test"),console.log("test2");

// if (balance < 500) {
//       console.log("less than 500");
// } else if (balance < 750) {
//       console.log("less than 750");

// } else if (balance < 900) {
//       console.log("less than 750");

// } else {
//       console.log("less than 1200");

// }

const userLoggedIn = true
const debitCard = true
const loggedInFromGoogle = false
const loggedInFromEmail = true

if (userLoggedIn && debitCard && 2==3) {
      console.log("Allow to buy course");
}

if (loggedInFromGoogle || loggedInFromEmail) {
      console.log("User logged in");
```

```
}

// switch (key) {
//         case value:

//                 break;

//         default:
//                 break;
// }

const month = "march"

switch (month) {
        case "jan":
                console.log("January");
                break;
        case "feb":
                console.log("feb");
                break;
        case "march":
                console.log("march");
                break;
        case "april":
                console.log("april");
                break;

        default:
                console.log("default case match");
                break;
}
```

```
const coding = ["js", "ruby", "java", "python", "cpp"]

// coding.forEach( function (val){
//         console.log(val);
// } )

// coding.forEach( (item) => {
//         console.log(item);
// } )

// function printMe(item){
//         console.log(item);
// }

// coding.forEach(printMe)

// coding.forEach( (item, index, arr)=> {
//         console.log(item, index, arr);
// } )

const myCoding = [
    {
        languageName: "javascript",
        languageFileName: "js"
    },
    {
        languageName: "java",
        languageFileName: "java"
    },
    {
        languageName: "python",
        languageFileName: "py"
```

```
    },
]

myCoding.forEach( (item) => {

    console.log(item.languageName);
} )


const myObject = {
    js: 'javascript',
    cpp: 'C++',
    rb: "ruby",
    swift: "swift by apple"
}

for (const key in myObject) {
    //console.log(`${key} shortcut is for ${myObject[key]}`);
}

const programming = ["js", "rb", "py", "java", "cpp"]

for (const key in programming) {
    //console.log(programming[key]);
}

// const map = new Map()
// map.set('IN', "India")
// map.set('USA', "United States of America")
// map.set('Fr', "France")
// map.set('IN', "India")
```

```
// for (const key in map) {
//     console.log(key);
// }


const myNums = [1, 2, 3]

// const myTotal = myNums.reduce(function (acc, currval) {
//     console.log(`acc: ${acc} and currval: ${currval}`);
//     return acc + currval
// }, 0)

const myTotal = myNums.reduce( (acc, curr) => acc+curr, 0)

console.log(myTotal);


const shoppingCart = [
    {
        itemName: "js course",
        price: 2999
    },
    {
        itemName: "py course",
        price: 999
    },
    {
        itemName: "mobile dev course",
        price: 5999
    },
    {
        itemName: "data science course",
```

```
        price: 12999
    },
]

const priceToPay = shoppingCart.reduce((acc, item) => acc + item.price, 0)

console.log(priceToPay);


// for

for (let i = 0; i <= 10; i++) {
    const element = i;
    if (element == 5) {
        //console.log("5 is best number");
    }
    //console.log(element);

}

// console.log(element);

for (let i = 1; i <= 10; i++) {
    //console.log(`Outer loop value: ${i}`);
   for (let j = 1; j <= 10; j++) {
    //console.log(`Inner loop value ${j} and inner loop ${i}`);
    //console.log(i + '*' + j + ' = ' + i*j );
   }

}
let myArray = ["flash", "batman", "superman"]
//console.log(myArray.length);
```

```
for (let index = 0; index < myArray.length; index++) {
    const element = myArray[index];
    //console.log(element);

}


// break and continue

// for (let index = 1; index <= 20; index++) {
//     if (index == 5) {
//         console.log(`Detected 5`);
//         break
//     }
//   console.log(`Value of i is ${index}`);

// }

for (let index = 1; index <= 20; index++) {
    if (index == 5) {
        console.log(`Detected 5`);
        continue
    }
  console.log(`Value of i is ${index}`);

}


const myNumers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

// const newNums = myNumers.map( (num) => { return num + 10})
```

```
const newNums = myNumers
                    .map((num) => num * 10 )
                    .map( (num) => num + 1)
                    .filter( (num) => num >= 40)

console.log(newNums);


// const coding = ["js", "ruby", "java", "python", "cpp"]


// const values = coding.forEach( (item) => {
//       //console.log(item);
//       return item
// } )

// console.log(values);

const myNums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

// const newNums = myNums.filter( (num) => {
//       return num > 4
// } )

// const newNums = []

// myNums.forEach( (num) => {
//       if (num > 4) {
//             newNums.push(num)
//       }
// } )
```

```
// console.log(newNums);


const books = [
    { title: 'Book One', genre: 'Fiction', publish: 1981, edition: 2004 },
    { title: 'Book Two', genre: 'Non-Fiction', publish: 1992, edition: 2008 },
    { title: 'Book Three', genre: 'History', publish: 1999, edition: 2007 },
    { title: 'Book Four', genre: 'Non-Fiction', publish: 1989, edition: 2010 },
    { title: 'Book Five', genre: 'Science', publish: 2009, edition: 2014 },
    { title: 'Book Six', genre: 'Fiction', publish: 1987, edition: 2010 },
    { title: 'Book Seven', genre: 'History', publish: 1986, edition: 1996 },
    { title: 'Book Eight', genre: 'Science', publish: 2011, edition: 2016 },
    { title: 'Book Nine', genre: 'Non-Fiction', publish: 1981, edition: 1989 },
  ];

  let userBooks = books.filter( (bk) => bk.genre === 'History')

  userBooks = books.filter( (bk) => {
    return bk.publish >= 1995 && bk.genre === "History"
})
  console.log(userBooks);


  // for of

// ["", "", ""]
// [{}, {}, {}]

const arr = [1, 2, 3, 4, 5]

for (const num of arr) {
    //console.log(num);
```

```
}

const greetings = "Hello world!"
for (const greet of greetings) {
    //console.log(`Each char is ${greet}`)
}

// Maps

const map = new Map()
map.set('IN', "India")
map.set('USA', "United States of America")
map.set('Fr', "France")
map.set('IN', "India")


// console.log(map);

for (const [key, value] of map) {
    // console.log(key, ':-', value);
}

const myObject = {
    game1: 'NFS',
    game2: 'Spiderman'
}

// for (const [key, value] of myObject) {
//     console.log(key, ':-', value);

// }
```

```
let index = 0
// while (index <= 10) {
//       console.log(`Value of index is ${index}`);
//       index = index + 2
// }

let myArray = ['flash', "batman", "superman"]

let arr = 0
while (arr < myArray.length) {
      //console.log(`Value is ${myArray[arr]}`);
      arr = arr + 1
}

let score = 11

do {
      console.log(`Score is ${score}`);
      score++
} while (score <= 10);



      function addLanguage(langName){
            const li = document.createElement('li');
            li.innerHTML = `${langName}`
            document.querySelector('.language').appendChild(li)
      }
      addLanguage("python")
      addLanguage("typescript")
```

```
function addOptiLanguage(langName){
    const li = document.createElement('li');
    li.appendChild(document.createTextNode(langName))
    document.querySelector('.language').appendChild(li)
}
addOptiLanguage('golang')

//Edit
const secondLang = document.querySelector("li:nth-child(2)")
console.log(secondLang);
//secondLang.innerHTML = "Mojo"
const newli = document.createElement('li')
newli.textContent = "Mojo"
secondLang.replaceWith(newli)

//edit
const firstLang = document.querySelector("li:first-child")
firstLang.outerHTML = '<li>TypeScript</li>'

//remove
const lastLang = document.querySelector('li:last-child')
lastLang.remove()
```